

Hands-on Lab: Working with Multiple Tables in MySQL using phpMyAdmin

Estimated time needed: 20 minutes

In this lab, you will learn how to create tables and load data in the MySQL database service using the phpMyAdmin graphical user interface (GUI) tool.

Software Used in this Lab

In this lab, you will use [MySQL](#). MySQL is a Relational Database Management System (RDBMS) designed to efficiently store, manipulate, and retrieve data.



To complete this lab you will utilize MySQL relational database service available as part of IBM Skills Network Labs (SN Labs) Cloud IDE. SN Labs is a virtual lab environment used in this course.

Database Used in this Lab

The database used in this lab is an internal database. You will be working on a sample HR database. This HR database schema consists of 5 tables called **EMPLOYEES**, **JOB_HISTORY**, **JOBS**, **DEPARTMENTS** and **LOCATIONS**. Each table has a few rows of sample data. The following diagram shows the tables for the HR database:

|

SAMPLE HR DATABASE TABLES											
EMPLOYEES											
EMP_ID	LAST_NAME	FIRST_NAME	DOB	DATE	SEX	ADDRESS	JOB_ID	SALARY	MANAGER_ID	DEPT_ID	
E1001	John	Thomas	123456	1976-01-09	M	5613 Rex, Oak Park A.	100	100000	30001	2	
E1002	Alice	James	123457	1972-07-31	F	980 Berry Ln, High/L	200	80000	30002	5	
E1003	Steve	Wells	123458	1980-08-10	M	251 Sempas, Grnys/L	300	50000	30002	5	

JOB_HISTORY				JOBS			
EMP_ID	START_DATE	JOB_ID	DEPT_ID	JOB_IDENT	JOB_TITLE	MIN_SALARY	MAX_SALARY
E1001	2000-01-30	100	2	100	Sr Architect	60000	100000
E1002	2010-08-16	200	5	200	Sr Software Developer	60000	80000
E1003	2010-08-10	300	5	300	Jr Software Developer	40000	60000

DEPARTMENTS				LOCATIONS	
DEPT_ID	DEPT_NAME	MANAGER_ID	LOC_ID	LOCATION	EMP_ID-LOC
2	Architect Group	30001	L0001	L0001	2
5	Software Development	30002	L0002	L0002	5
7	Design Team	30003	L0003	L0003	7

Objectives

After completing this lab you will be able to:

- Write SQL queries that access more than one table
- Compose queries that access multiple tables using a nested statement in the WHERE clause
- Build queries with multiple tables in the FROM clause
- Write Implicit Join queries with join criteria specified in the WHERE clause
- Specify aliases for table names and qualify column names with table aliases

In this lab, you will through some SQL practice problems that will provide hands-on experience with SQL queries that access multiple tables. You will be:

- Accessing Multiple Tables with Sub-Queries
- Accessing Multiple Tables with Implicit Joins

How does an Implicit version of CROSS JOIN (also known as Cartesian Join) statement syntax look?

```
SELECT column_name(s)
FROM table1, table2;
```

How does an Implicit version of INNER JOIN statement syntax look?

```
SELECT column_name(s)
FROM table1, table2
WHERE table1.column_name = table2.column_name;
```

Exercise 1: Accessing Multiple Tables with Sub-Queries

1. Problem:

Retrieve only the EMPLOYEES records that correspond to jobs in the JOBS table.

► Solution

```
select * from EMPLOYEES where JOB_ID IN (select JOB_IDENT from JOBS);
```

► Output

+ Options

←

T

→

EMP_ID

F_NAME

L_NAME

SSN

B_DATE

SEX

ADDRESS

JOB_ID

SALARY

MANAGER_ID

DEP_ID

☐				E1001	John	Thomas	123456	1978-09-01	M	5631 Rice, OakPark,IL	100	100000.00	30001	2
☐				E1002	Alice	James	123457	1972-07-31	F	980 Berry In, Elgin,IL	200	80000.00	30002	5
☐				E1003	Steve	Wells	123458	1980-10-08	M	291 Springs, Gary,IL	300	50000.00	30002	5
☐				E1004	Sanjosh	Kumar	123459	1985-07-20	M	511 Aurora Av, Aurora,IL	400	60000.00	30004	5
☐				E1005	Ahmed	Hussain	123410	1981-04-01	M	216 Oak Tree, Geneva,IL	500	70000.00	30001	2
☐				E1006	Nancy	Allen	123411	1978-06-02	F	111 Green Pl, Elgin,IL	600	90000.00	30001	2
☐				E1007	Mary	Thomas	123412	1975-05-05	F	100 Rose Pl, Gary,IL	650	65000.00	30003	7
☐				E1008	Bharath	Gupta	123413	1985-06-05	M	145 Berry Ln, Naperville,IL	660	65000.00	30003	7
☐				E1009	Andrea	Jones	123414	1990-09-07	F	120 Fall Creek, Gary,IL	234	70000.00	30003	7
☐				E1010	Ann	Jacob	123415	1982-03-30	F	111 Britany Springs,Elgin,IL	220	70000.00	30004	5

2. Problem:

Retrieve only the list of employees whose JOB_TITLE is Jr. Designer:

Solution

```
select * from EMPLOYEES where JOB_ID IN (select JOB_IDENT from JOBS where JOB_TITLE= 'Jr. Designer');
```

Output

+ Options

←

T

→

JOB_TITLE

MIN_SALARY

MAX_SALARY

JOB_IDENT

☐				E1007	Mary	Thomas	123412	1975-05-05	F	100 Rose Pl, Gary,IL	650	65000.00	30003	7
☐				E1008	Bharath	Gupta	123413	1985-06-05	M	145 Berry Ln, Naperville,IL	660	65000.00	30003	7

↑

☐ Check all

With selected:

Edit

Copy

Delete

Export

3. Problem:

Retrieve JOB information and who earn more than \$70,000.

Solution

```
select JOB_TITLE, MIN_SALARY,MAX_SALARY,JOB_IDENT from JOBS where JOB_IDENT IN (select JOB_ID from EMPLOYEES where SALARY > 70000 );
```

Output

☐ Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

+ Options

←

T

→

JOB_TITLE

MIN_SALARY

MAX_SALARY

JOB_IDENT

☐				Sr Architect	60000.00	100000.00	100
☐				Sr Software Developer	60000.00	80000.00	200
☐				Lead Architect	70000.00	100000.00	600

↑

☐ Check all

With selected:

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Export

4. Problem:

Retrieve JOB information and list of employees whose birth year is after 1976.

Solution

```
select JOB_TITLE, MIN_SALARY,MAX_SALARY,JOB_IDENT from JOBS where JOB_IDENT IN (select JOB_ID from EMPLOYEES where YEAR(B_DATE)>1976 );
```

Output

+ Options

←

T

→

JOB_TITLE

MIN_SALARY

MAX_SALARY

JOB_IDENT

☐				Jr Software Developer	40000.00	60000.00	300
☐				Jr Software Developer	40000.00	60000.00	400
☐				Jr Architect	50000.00	70000.00	500
☐				Lead Architect	70000.00	100000.00	600
☐				Jr Designer	60000.00	70000.00	660
☐				Sr Designer	70000.00	90000.00	234
☐				Sr Designer	70000.00	90000.00	220

↑

☐ Check all

With selected:

Edit

Copy

Delete

Export

5. Problem:

Retrieve JOB information and list of female employees whose birth year is after 1976.

► Solution

```
select JOB_TITLE, MIN_SALARY, MAX_SALARY, JOB_IDENT from JOBS where JOB_IDENT IN (select JOB_ID from EMPLOYEES where YEAR(B_DATE)>1976 and SEX='F' );
```

► Output

+ Options

				JOB_TITLE	MIN_SALARY	MAX_SALARY	JOB_IDENT
☐				Lead Architect	70000.00	100000.00	600
☐				Sr. Designer	70000.00	90000.00	234
☐				Sr. Designer	70000.00	90000.00	220

Exercise 2: Accessing Multiple Tables with Implicit Joins

1. Problem:

Perform an implicit cartesian/cross join between EMPLOYEES and JOBS tables.

► Solution

```
select * from EMPLOYEES, JOBS;
```

► Output

+ Options

EMP_ID	F_NAME	L_NAME	SSN	B_DATE	SEX	ADDRESS	JOB_ID	SALARY	MANAGER_ID	DEP_ID	JOB_IDENT	JOB_TITLE	MIN_SALARY	MAX_SALARY
E1010	Ann	Jacob	123415	1982-03-30	F	111 Britnary Springs, Elgin, IL	220	70000.00	30004	5	100	Sr. Architect	60000.00	100000.00
E1009	Andrea	Jones	123414	1990-09-07	F	120 Fall Creek, Gary, IL	234	70000.00	30003	7	100	Sr. Architect	60000.00	100000.00
E1008	Bharath	Gupta	123413	1985-06-05	M	145 Berry Ln, Naperville, IL	660	65000.00	30003	7	100	Sr. Architect	60000.00	100000.00
E1007	Mary	Thomas	123412	1975-05-05	F	100 Rose Pl, Gary, IL	650	65000.00	30003	7	100	Sr. Architect	60000.00	100000.00
E1006	Nancy	Allen	123411	1978-06-02	F	111 Green Pl, Elgin, IL	600	90000.00	30001	2	100	Sr. Architect	60000.00	100000.00
E1005	Ahmed	Hussain	123410	1981-04-01	M	216 Oak Tree, Geneva, IL	500	70000.00	30001	2	100	Sr. Architect	60000.00	100000.00
E1004	Santosh	Kumar	123459	1985-07-20	M	511 Aurora Av, Aurora, IL	400	60000.00	30004	5	100	Sr. Architect	60000.00	100000.00
E1003	Steve	Wells	123458	1980-10-08	M	291 Springs, Gary, IL	300	50000.00	30002	5	100	Sr. Architect	60000.00	100000.00
E1002	Alice	James	123457	1972-07-31	F	980 Berry In, Elgin, IL	200	80000.00	30002	5	100	Sr. Architect	60000.00	100000.00
E1001	John	Thomas	123456	1976-09-01	M	5631 Rice, OakPark, IL	100	100000.00	30001	2	100	Sr. Architect	60000.00	100000.00
E1010	Ann	Jacob	123415	1982-03-30	F	111 Britnary Springs, Elgin, IL	220	70000.00	30004	5	200	Sr. Software Developer	60000.00	80000.00
E1009	Andrea	Jones	123414	1990-09-07	F	120 Fall Creek, Gary, IL	234	70000.00	30003	7	200	Sr. Software Developer	60000.00	80000.00
E1008	Bharath	Gupta	123413	1985-06-05	M	145 Berry Ln, Naperville, IL	660	65000.00	30003	7	200	Sr. Software Developer	60000.00	80000.00
E1007	Mary	Thomas	123412	1975-05-05	F	100 Rose Pl, Gary, IL	650	65000.00	30003	7	200	Sr. Software Developer	60000.00	80000.00
E1006	Nancy	Allen	123411	1978-06-02	F	111 Green Pl, Elgin, IL	600	90000.00	30001	2	200	Sr. Software Developer	60000.00	80000.00

2. Problem:

Retrieve only the EMPLOYEES records that correspond to jobs in the JOBS table.

► Solution

```
select * from EMPLOYEES, JOBS where EMPLOYEES.JOB_ID = JOBS.JOB_IDENT;
```

► Output

Showing rows 0 - 9 (10 total, Query took 0.0006 seconds.)

```
select * from EMPLOYEES, JOBS where EMPLOYEES.JOB_ID = JOBS.JOB_ID
```

Profiling

Edit inline

Edit

Explain SQL

Create PHP code

Refresh

Show all

Number of rows: 25

Filter rows: Search this table

+ Options

EMP_ID	F_NAME	L_NAME	SSN	B_DATE	SEX	ADDRESS	JOB_ID	SALARY	MANAGER_ID	DEP_ID	JOB_IDENT	JOB_TITLE	MIN_SALARY	MAX_SALARY
E1001	John	Thomas	123456	1976-09-01	M	5631 Rice, OakPark,IL	100	100000.00	30001	2	100	Sr. Architect	60000.00	100000.00
E1002	Alice	James	123457	1972-07-31	F	980 Berry Ln, Elgin,IL	200	80000.00	30002	5	200	Sr Software Developer	60000.00	80000.00
E1003	Steve	Wells	123458	1980-10-08	M	291 Springs, Gary,IL	300	50000.00	30002	5	300	Jr Software Developer	40000.00	60000.00
E1004	Santosh	Kumar	123459	1985-07-20	M	511 Aurora Av, Aurora,IL	400	60000.00	30004	5	400	Jr Software Developer	40000.00	60000.00
E1005	Ahmed	Hussain	123410	1981-04-01	M	216 Oak Tree, Geneva,IL	500	70000.00	30001	2	500	Jr. Architect	50000.00	70000.00
E1006	Nancy	Allen	123411	1978-06-02	F	111 Green Pl, Elgin,IL	600	90000.00	30001	2	600	Lead Architect	70000.00	100000.00
E1007	Mary	Thomas	123412	1975-05-05	F	100 Rose Pl, Gary,IL	650	65000.00	30003	7	650	Jr. Designer	60000.00	70000.00
E1008	Bharath	Gupta	123413	1985-06-05	M	145 Berry Ln, Naperville,IL	660	65000.00	30003	7	660	Jr. Designer	60000.00	70000.00
E1009	Andrea	Jones	123414	1990-09-07	F	120 Fall Creek, Gary,IL	234	70000.00	30003	7	234	Sr. Desioner	70000.00	90000.00

3. Problem:

Redo the previous query, using shorter aliases for table names.

Solution

```
select * from EMPLOYEES E, JOBS J where E.JOB_ID = J.JOB_ID;
```

Showing rows 0 - 9 (10 total, Query took 0.0008 seconds.)

```
select * from EMPLOYEES E, JOBS J where E.JOB_ID = J.JOB_ID
```

Profiling

Edit inline

Edit

Explain SQL

Create PHP code

Refresh

Show all

Number of rows: 25

Filter rows: Search this table

+ Options

EMP_ID	F_NAME	L_NAME	SSN	B_DATE	SEX	ADDRESS	JOB_ID	SALARY	MANAGER_ID	DEP_ID	JOB_IDENT	JOB_TITLE	MIN_SALARY	MAX_SALARY
E1001	John	Thomas	123456	1976-09-01	M	5631 Rice, OakPark,IL	100	100000.00	30001	2	100	Sr. Architect	60000.00	100000.00
E1002	Alice	James	123457	1972-07-31	F	980 Berry Ln, Elgin,IL	200	80000.00	30002	5	200	Sr Software Developer	60000.00	80000.00
E1003	Steve	Wells	123458	1980-10-08	M	291 Springs, Gary,IL	300	50000.00	30002	5	300	Jr Software Developer	40000.00	60000.00
E1004	Santosh	Kumar	123459	1985-07-20	M	511 Aurora Av, Aurora,IL	400	60000.00	30004	5	400	Jr Software Developer	40000.00	60000.00
E1005	Ahmed	Hussain	123410	1981-04-01	M	216 Oak Tree, Geneva,IL	500	70000.00	30001	2	500	Jr. Architect	50000.00	70000.00
E1006	Nancy	Allen	123411	1978-06-02	F	111 Green Pl, Elgin,IL	600	90000.00	30001	2	600	Lead Architect	70000.00	100000.00
E1007	Mary	Thomas	123412	1975-05-05	F	100 Rose Pl, Gary,IL	650	65000.00	30003	7	650	Jr. Designer	60000.00	70000.00
E1008	Bharath	Gupta	123413	1985-06-05	M	145 Berry Ln, Naperville,IL	660	65000.00	30003	7	660	Jr. Designer	60000.00	70000.00
E1009	Andrea	Jones	123414	1990-09-07	F	120 Fall Creek, Gary,IL	234	70000.00	30003	7	234	Sr. Designer	70000.00	90000.00

4. Problem:

Redo the previous query, but retrieve only the Employee ID, Employee Name and Job Title.

Solution

```
select EMP_ID,F_NAME,L_NAME, JOB_TITLE from EMPLOYEES E, JOBS J where E.JOB_ID = J.JOB_ID;
```

Output

Showing rows 0 - 9 (10 total, Query took 0.0006 seconds)

```
select EMP_ID,F_NAME,L_NAME, JOB_TITLE from EMPLOYEES E, JOBS J where E.JOB_ID = J.JOB_IDENT
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all

Number of rows: 25

Filter rows: Search this table

+ Options

EMP_ID	F_NAME	L_NAME	JOB_TITLE
E1001	John	Thomas	Sr. Architect
E1002	Alice	James	Sr. Software Developer
E1003	Steve	Wells	Jr. Software Developer
E1004	Santosh	Kumar	Jr. Software Developer
E1005	Ahmed	Hussain	Jr. Architect
E1006	Nancy	Allen	Lead Architect
E1007	Mary	Thomas	Jr. Designer
E1008	Bharath	Gupta	Jr. Designer
E1009	Andrea	Jones	Sr. Designer
E1010	Ann	Jacob	Sr. Designer

Show all

Number of rows: 25

Filter rows: Search this table

5. Problem:

Redo the previous query, but specify the fully qualified column names with aliases in the SELECT clause.

► Solution

```
select E.EMP_ID,E.F_NAME,E.L_NAME, J.JOB_TITLE from EMPLOYEES E, JOBS J where E.JOB_ID = J.JOB_IDENT;
```

Showing rows 0 - 9 (10 total, Query took 0.0010 seconds)

```
select E.EMP_ID,E.F_NAME,E.L_NAME, J.JOB_TITLE from EMPLOYEES E, JOBS J where E.JOB_ID = J.JOB_IDENT
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all

Number of rows: 25

Filter rows: Search this table

+ Options

EMP_ID	F_NAME	L_NAME	JOB_TITLE
E1001	John	Thomas	Sr. Architect
E1002	Alice	James	Sr. Software Developer
E1003	Steve	Wells	Jr. Software Developer
E1004	Santosh	Kumar	Jr. Software Developer
E1005	Ahmed	Hussain	Jr. Architect
E1006	Nancy	Allen	Lead Architect
E1007	Mary	Thomas	Jr. Designer
E1008	Bharath	Gupta	Jr. Designer
E1009	Andrea	Jones	Sr. Designer
E1010	Ann	Jacob	Sr. Designer

Show all

Number of rows: 25

Filter rows: Search this table

Solution Script

If you would like to run all the solution queries of the SQL problems of this lab with a script, download the script below. Import the script to mysql phpadmin interface and run. Follow [Hands-on Lab : Create tables using SQL scripts and Load data into tables](#) on how to import a script to MYsql phpadmin interface and run it.

- [MultipleTables_Solution_Script.sql](#)

Congratulations! You have completed this lab, and you are ready for the next topic.

Author(s)

[Lakshmi Holla](#)
[Malika Sinha](#)

